

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Maschinelles Lernen

Exam: IN2064 / Retake
Examiner: Prof. Günnemann

Date: Tuesday 11th April, 2023
Time: 16:30 – 18:30

	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9
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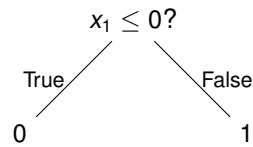
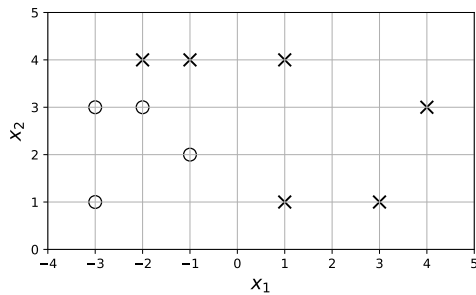
Working instructions

- This exam consists of **12 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 41 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____

Problem 1 Decision trees (5 credits)

Consider the following two-dimensional classification dataset with classes “0” (o marker) and 1 (x marker) and the corresponding decision tree:



- 0 ☐ a) Calculate the Gini index $i_G(t_{\text{root}})$ of the root node, $i_G(t_{x_1 \leq 0})$ of the left node, and $i_G(t_{x_1 > 0})$ of the right node. Simplify your results to a single fraction, i.e. $\frac{a}{b}$.

1 ☐

2 ☐

3 ☐

- 0 ☐ b) Calculate the improvement Δi_G of the Gini index caused by the first split. Simplify your results to a single fraction, i.e. $\frac{a}{b}$.

1 ☐

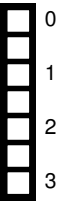
2 ☐

Problem 2 Probabilistic inference (6 credits)

a) You are given N values $x_1, \dots, x_N$ that were independently sampled from uniform distribution $U(0, b)$ with $b \in [0, 1]$ and density function

$$p(x | b) = \begin{cases} \frac{1}{b} & \text{if } 0 \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Derive the maximum likelihood estimate $b_{\text{MLE}} = \operatorname{argmax}_{b \in [0, 1]} \prod_{n=1}^N p(x_n | b)$.



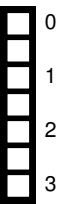
b) Now assume that b is a random variable with $b \sim \text{Beta}(\alpha, 1)$, i.e.

$$p(b) = \begin{cases} \frac{1}{c} \cdot b^{\alpha-1} & \text{if } 0 \leq b \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

where $c \in \mathbb{R}_+$ is a normalizing constant and $\alpha > 0$.

Derive the maximum a posteriori estimate $b_{\text{MAP}} = \operatorname{argmax}_{b \in [0, 1]} p(b | x_1, \dots, x_N)$.

Note: You will have to make a case distinction based on the value of α .



Problem 3 Linear regression (6 credits)

We want to perform regularized linear regression (without bias) on a dataset with N samples $\mathbf{x}_i \in \mathbb{R}^d$ with corresponding targets y_i (represented compactly as $\mathbf{X} \in \mathbb{R}^{N \times d}$ and $\mathbf{y} \in \mathbb{R}^N$). You assume that your targets stem from a generalized Gaussian distribution with parameters $\alpha, \beta > 0$, i.e.,

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = \frac{\beta}{2\alpha\Gamma(1/\beta)} \exp\left(-\left(\frac{|y_i - \mathbf{x}_i^T \mathbf{w}|}{\alpha}\right)^\beta\right) \quad (3.1)$$

with $\Gamma : \mathbb{R}_+ \mapsto \mathbb{R}_+$ being the Gamma function:

$$\Gamma(x) = \int_0^\infty t^{x-1} \exp(-t) dt. \quad (3.2)$$

0 ☐ a) **Derive** the negative logarithm of the likelihood, i.e., $-\log p(\mathbf{y} | \mathbf{X}, \mathbf{w})$ (up to some additive constant w.r.t. $\mathbf{w}$).

1 ☐

2 ☐

3 ☐

0 ☐ b) To add regularization, you add the logistic distribution as prior on your parameters

1 ☐

2 ☐

3 ☐

$$p(\mathbf{w}) = \prod_{i=1}^d \frac{\exp\left(-\frac{w_i}{s}\right)}{s(1 + \exp(-\frac{w_i}{s}))^2} \quad (3.6)$$

with parameter $s > 0$. **Derive** the negative logarithm of the posterior, i.e., $-\log p(\mathbf{w} | \mathbf{X}, \mathbf{y})$ (up to some additive constant w.r.t. $\mathbf{w}$).

Problem 4 Classification (4 credits)

You are given a dataset with $N = 4$ samples $\mathbf{x} \in [0, 1] \times \mathbb{R}$ and a naive Bayes classifier with classes $y \in \{1, 2\}$ with the prior

$$p(y = k) = \begin{cases} 0.7 & \text{if } k = 1 \\ 0.3 & \text{if } k = 2 \end{cases} \quad (4.1)$$

and feature distributions

$$p(x_1|y = k) = \begin{cases} \text{Bernoulli}(x_1|1) & \text{if } k = 1 \\ \text{Bernoulli}(x_1|0.7) & \text{if } k = 2 \end{cases}, \quad (4.2)$$

$$p(x_2|y = k) = \begin{cases} U(x_2|0, 5) & \text{if } k = 1 \\ U(x_2|0, 1) & \text{if } k = 2 \end{cases} \quad (4.3)$$

where $Bernoulli(x|p) = p$ is the Bernoulli distribution and

$$U(x|a, b) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a, b) \\ 0 & \text{else} \end{cases} \quad (4.4)$$

is the uniform distribution.

Classify the following four data points:

	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$
x_1	1	1	1	0
x_2	0.7	3.5	1.0	0.4



Problem 5 Optimization (4 credits)

Below, you are given two different functions and asked to **prove** convexity.

0	☐
1	☐
2	☐
3	☐
4	☐

You are given the *quadratic-over-linear* function

$$f(x, y) = \frac{x^2}{y}$$

Prove $f(x, y)$ is convex on the domain $\mathcal{X} = \mathbb{R} \times \mathbb{R}_{>0} = \{(x, y) \in \mathbb{R}^2 | y > 0\}$.

Hint: Remember that a (symmetric) matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is positive semi-definitive, if $\forall \mathbf{z} \in \mathbb{R}^d : \mathbf{z}^\top \mathbf{A} \mathbf{z} \geq 0$

Problem 6 Deep Learning (4 credits)

Layer normalization improves training stability by normalizing the input to have zero mean and unit standard deviation (similarly to BatchNorm). Layer normalization over the batch dimension takes as input a vector $x \in \mathbb{R}^m$ and transforms it as

$$\mu = \frac{1}{m} \sum_{i=1}^m x_i \quad (6.1)$$

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu)^2 \quad (6.2)$$

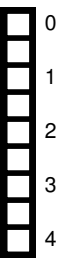
$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (6.3)$$

$$y_i = \gamma \hat{x}_i + \beta \quad (6.4)$$

$$(6.5)$$

In the following tasks, we want to compute the backward pass through this module. As input we get the gradient $\frac{d\mathcal{L}}{dy_i}$ w.r.t. the transformed output.

Derive the gradient of the loss w.r.t to the input x_i , that is, $\frac{d\mathcal{L}}{dx_i}$. First factorize the gradient according to the chain rule and then compute the individual factors.



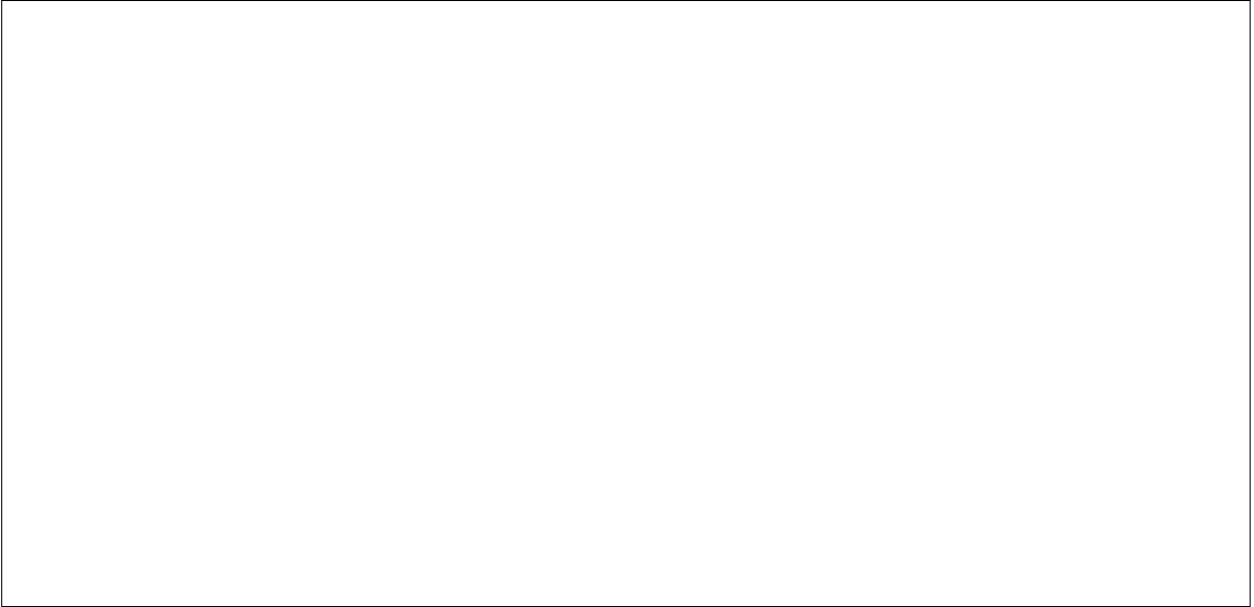
Problem 7 Kernels (6 credits)

0 ☐ a) Given vectors a, b and non-negative scalar c and $p \in \mathbb{N}$, **prove** that the polynomial kernel $k(a, b) = (a^T b + c)^p$ is a valid kernel.

1 ☐
2 ☐ **Hint:** You can use the kernel preserving operations from the lecture, given again as reference.
3 ☐ Let $k_1 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and $k_2 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be kernels, with $\mathcal{X} \subseteq \mathbb{R}^N$. Then the following functions k are kernels as well:

1. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) + k_2(\mathbf{x}_1, \mathbf{x}_2)$
2. $k(\mathbf{x}_1, \mathbf{x}_2) = c \cdot k_1(\mathbf{x}_1, \mathbf{x}_2)$, with $c > 0$
3. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) \cdot k_2(\mathbf{x}_1, \mathbf{x}_2)$
4. $k(\mathbf{x}_1, \mathbf{x}_2) = k_3(\phi(\mathbf{x}_1), \phi(\mathbf{x}_2))$, with the kernel k_3 on $\mathcal{X}' \subseteq \mathbb{R}^M$ and $\phi : \mathcal{X} \rightarrow \mathcal{X}'$
5. $k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^T \mathbf{A} \mathbf{x}_2$, with $\mathbf{A} \in \mathbb{R}^N \times \mathbb{R}^N$ symmetric and positive semidefinite

0 ☐ b) Assume a quadratic kernel $k(a, b) = (a^T b)^2$ with two-dimensional vectors $a, b \in \mathbb{R}^2$ as input. **Derive** a feature map $\phi(a)$ such that $k(a, b) = \phi(a)^T \phi(b)$.



Problem 8 Dimensionality reduction (3 credits)

0 ☐
1 ☐
2 ☐
3 ☐

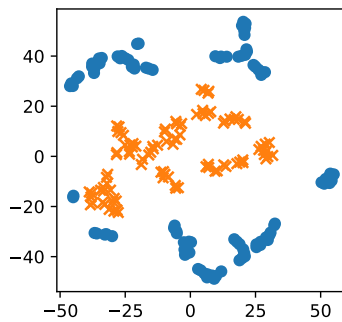
You are given a dataset $\mathbf{X}$ with 200 three-dimensional points. Half of points lie on a sphere with radius 1, and the other half lie on a sphere with radius 0.5, both with a center at origin. You are given routines for PCA as well as TSNE and use them to perform the following transformations:

- a) $\mathbf{A} = \text{PCA}(\mathbf{X}, \text{n_components}=2)$
- b) $\mathbf{B} = \text{TSNE}(\mathbf{X}, \text{n_components}=2, \text{perplexity}=5)$
- c) $\mathbf{C} = \text{TSNE}(\mathbf{X}, \text{n_components}=2, \text{perplexity}=100)$

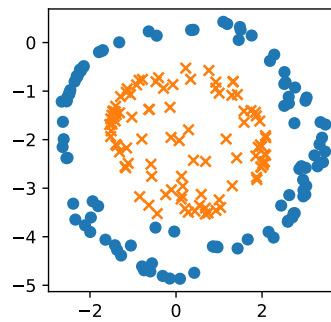
which gives you three down-projected matrices $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{200 \times 2}$.

Three figures below plot the projected data points using a)–c). Blue circles correspond to points which belong to a radius 1 sphere and orange crosses belong to points which belong to radius 0.5 sphere.

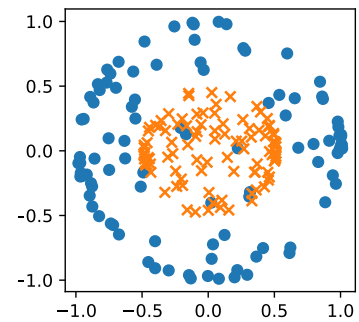
Pair a), b), c) with plots 1), 2), 3) below. Justify your answer.



1)



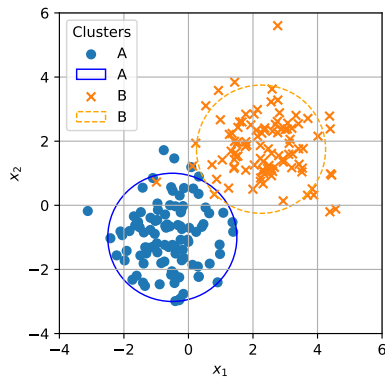
2)



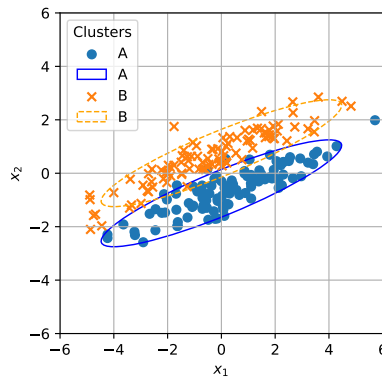
3)

Problem 9 Clustering (3 credits)

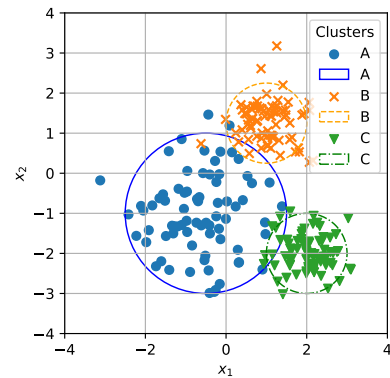
You are given three different two-dimensional datasets $\mathbf{X} \in \mathbb{R}^{n \times 2}$ with varying number of data points n . Dataset (a) and (b) consist of two clusters and (c) of three. The ground truth cluster assignments are given by color and marker style: blue circles, orange crosses, and green triangles (see legend). Each ground-truth cluster follows a multivariate Gaussian distribution. We plot the $4\text{-}\sigma$ covariance ellipses, i.e., each point on this ellipses has equal density (also known as probability contour or likelihood isobar).



(a)



(b)



(c)

a) What are the optimal decision boundaries for the k-means objective? With decision boundaries we refer to the hyperplanes that are of equal distance to the two closest cluster centroids (also known as Voronoi tessellation). In other words, the points separated by the decision boundaries.

Your task is to **add/sketch the optimal decision boundaries to/in the plots above**. That is, you applied a k-means clustering algorithm that found the globally optimal centroids and you choose k s.t. it matches the number of ground truth clusters.

0
1

b) For which dataset is it better to apply k-Means clustering and for which dataset Gaussian mixture models (GMM)? **Justify your answer.**

0
1

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

